SAP BI/BW Security

SAP BI/BW Synonymy called.

- ✓ BI/BW was introduced in the year **1998.**
- ✓ BW stands for **Business warehouse.**
- ✓ BI stands for **Business Intelligence.**

- BI/BW is nothing but **data warehousing tool of SAP.**
- **Data warehousing means** the tool, which is used for analysis of data, pulling of data, report execution, report forming, report preparation.
- For any business, what does the purpose of **data warehousing tools mean**?

= > The tool is used to generate **reports.**

= > What sort of reports mean daily expenditures, the reports related to revenue, the reports related to sales, profits, etc.
= > These reports once they are generated are used for analysis and accordingly, they plan for their business.

Ex = > non-SAP = **Informatica**, **Cognos**.

= > within SAP we use BI/BW system.

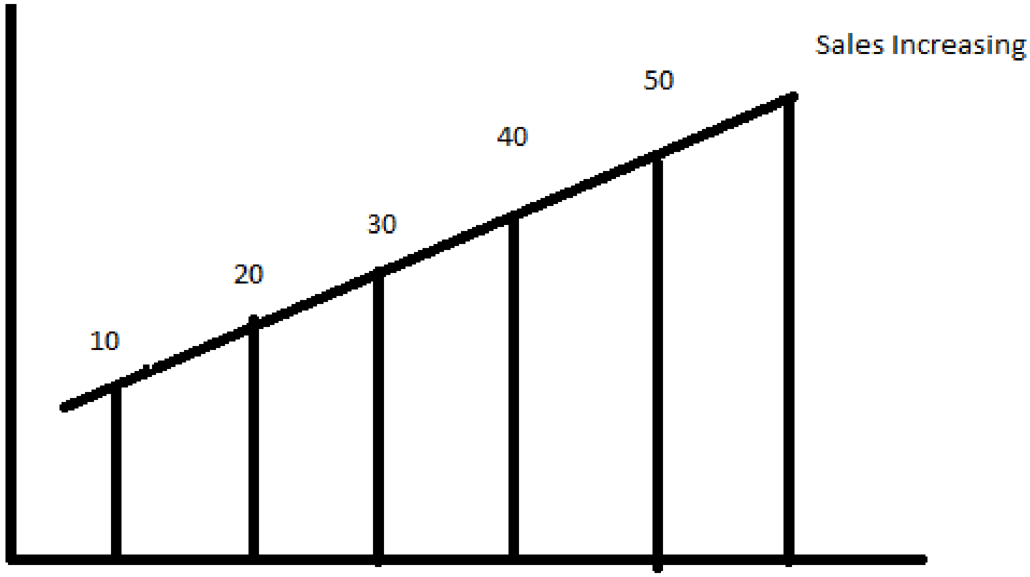
What Exactly the data warehousing tool is used for?

- In any business the main aim of the business is growth; in order to grow your business first you need to grow your sales and revenue.
- How do we grow means for ex = > In the automobile industry, Toyota, Hyundai, whenever they launch the new products, what basis they are launching, what are the factors they consider before the launching of a car.
- They definitely have to consider the aspects before the launch. So, what type of business or factors before launching, they have to gather the data, Income, range, rate etc.

For ex = > If we take sales area, assume any product is launched,

- In the sales area also it has to be increased accordingly to reach this point(in the diagram) you need to know the data for the previous year also, like month wise, yearly, if we know the last 10 years, 20 years data, how much it is **increased** and it is **decreased**, these we can easily achieve in the goal of sales.

- The usage of data warehousing is **increasing**; Hadoop big data is also growing a lot when compared to other **technologies.**



- If the data become increasing the data become **huge**, you need a tool definitely to control, or to analyze the data to prepare a report from data.
- These are the data sources from multiple resources the data is loading is called **source** and from monthly the data is loaded into **BI system.**

<u>Data Consolidation</u> = > meaning is **combined** or bring **together.**

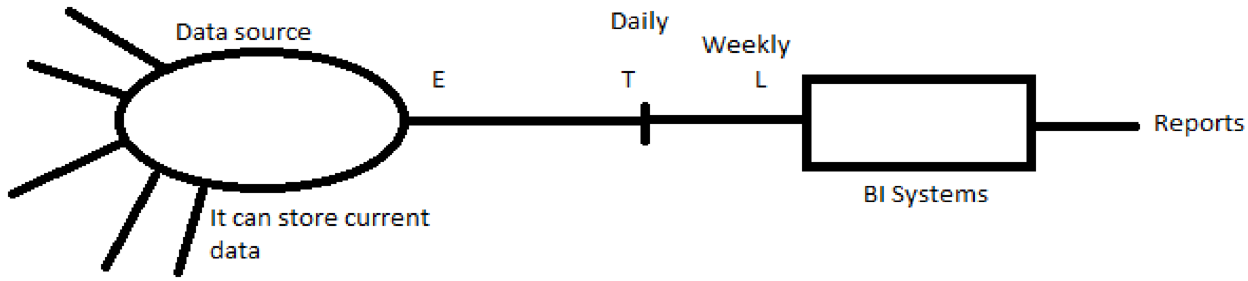
What is **data warehousing tool** does?

First it will **consolidate** the data, the consolidation meaning is the data is combined or bringing together.

Secondly the **Analysis**, Once the data is gathered into the source, we are analyzing the **data.**

The third one is **planning**, the outcome of the analysis is called planning or **forecasting**, after 50 years the sales have to be increased right, this is called planning.

<u>ETL Tool</u>



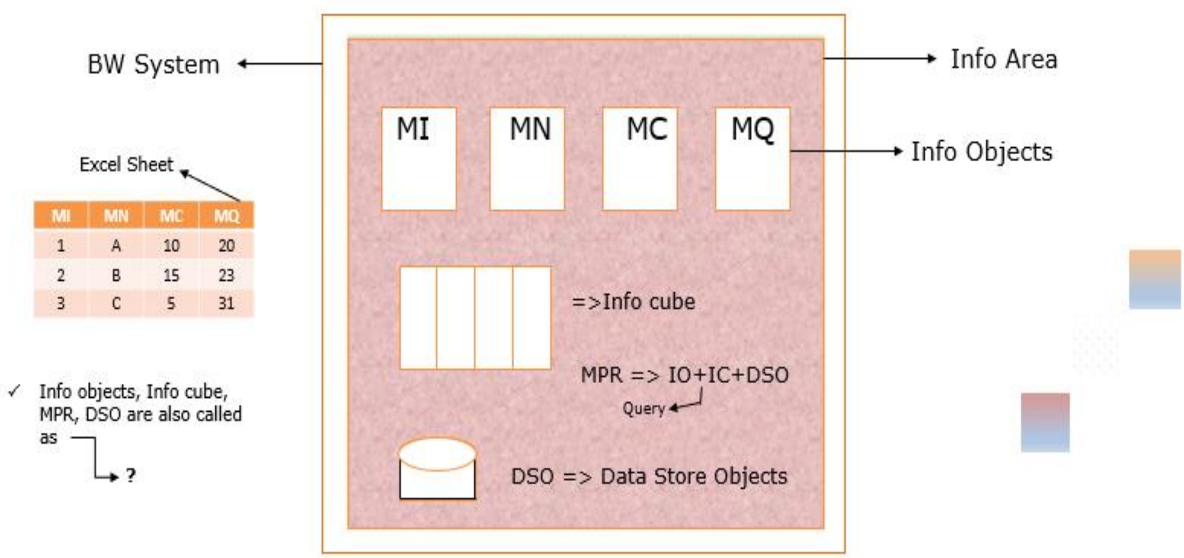
<u>**ETL stands for Extraction, Transformation, and Loading.**</u>

In the above **diagram** you can see the data is **extracted** from the **data source** and in the **middle** the data is **transformed**, and it is loaded into **BI systems**. loading can happen weekly, monthly, and on a daily basis and finally that can be extracted into the reports for the **business requirements.**

What exactly does **Transformation** mean?

For Ex = > In any **Oil Business,** the oil can be measured in terms of **barrels;** they sell the product, buy the product from country to country in tons of barrels only, but the system **does not know** what barrel is. but it has to be **transferred in liters.**

But how can the **Loading** be Done?

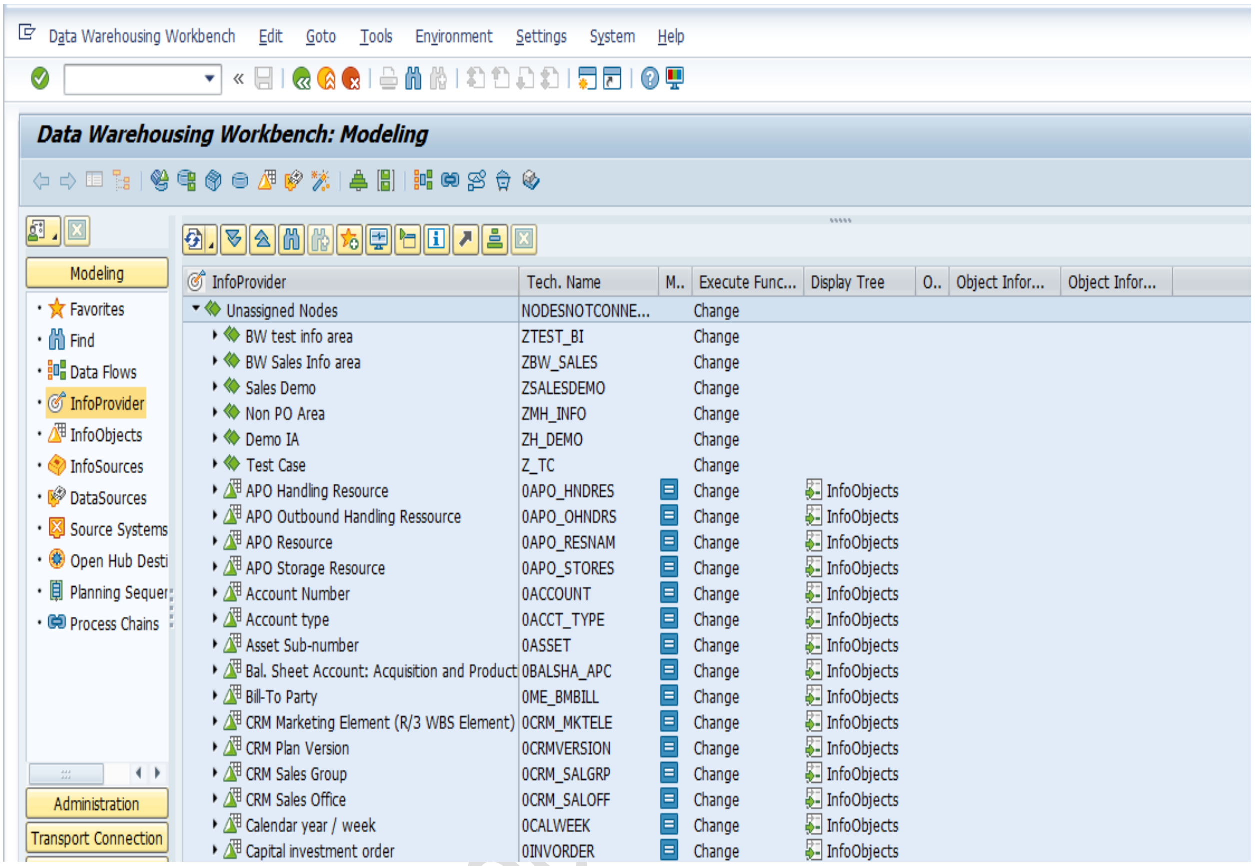


Let us take an **example**, in the above diagram, here you can see the simple excel sheet, the data **containing** excel sheet, but how these data are loading into **BI/BW systems means –**

- Whatever the **columns** are there like **MI, MN, MC, MQ** that could be called as **terminology** in BW Systems.
- In the BW System, the **BW team**, first they will create something called as **Info area,** so there is the outer box what we see that is called **BW System.** = > Info area is nothing but a **private space or independent space.**
- Inside the info area**, MI, MN, MC, MQ**, these are nothing but info objects, but what SAP recommends, if we create any new **info objects**, which could start with **Z.**
- Once the info objects are created, there will be a **mapping done**, from the data source (Excel sheet) to BI systems. The mapping will be done like MI – Material Id, MN – Material Name, Material Code – MQ, Material Quantity – MQ, **this is loading.**
- Once they start **loading,** all the column values that can come to store in the respective **info objects**, these info objects are grouped together or combined together to form an **info cube**.(but in reality, you can see in the systems **in 3 dimensional**).
- Inside the Info area, we can see one more diagram called **DSO**(data store objects).
- DSO stands for **data store objects**; it is also called an **independent storage device**. Even we can write the query on DSO, but it contains only the current data, hence we cannot write queries on DSO, but whereas info cube, multi providers contain **new data as well as past data.**
- **Multi providers** means = > It is a **combination** of info cube, info objects and DSO, because multi providers contain more data.
- === > These are all four types of **storage devices,** and they are also called Info providers. (IC+IO+DSO+MPR). = > Info cube + Info object + Data store objects + Multi providers.
- This information can be seen in the t code called **RSA1** (it stands for data warehouse workbench) where everything is done by starting from extraction, transformation and loading, report execution etc.
- Below is the screenshot of RSA1 T Code, when we click **on left side modelling options,** it will show the info objects,

Then go to the info provider (**double diamond symbol**), referring to an info area and its inside info object is there (a triangle with square mesh in the background).

Out of 4 options maximum **information** can contain multi provider or more data in it, the report or query written there only.



There is no use of **writing** query on individual info objects, they will write the query on info cube, info objects, and multi providers etc.

Types of **<u>Info providers</u>**(in simple words) = >

Info object = > above ex (Inside the info area, MI, MN, MC, MQ, these are nothing but info objects, but what SAP recommends, if we create any new info objects, that could start with Z)

Info cube = > it contains more data

DSO = > data store objects (even we can write the query on DSO, but it contains only current data)

Multi providers = > also contain more data queries/reports written in info cube or multi providers only.

<u>Query execution</u>

The T Codes related to this are **RRMX.**

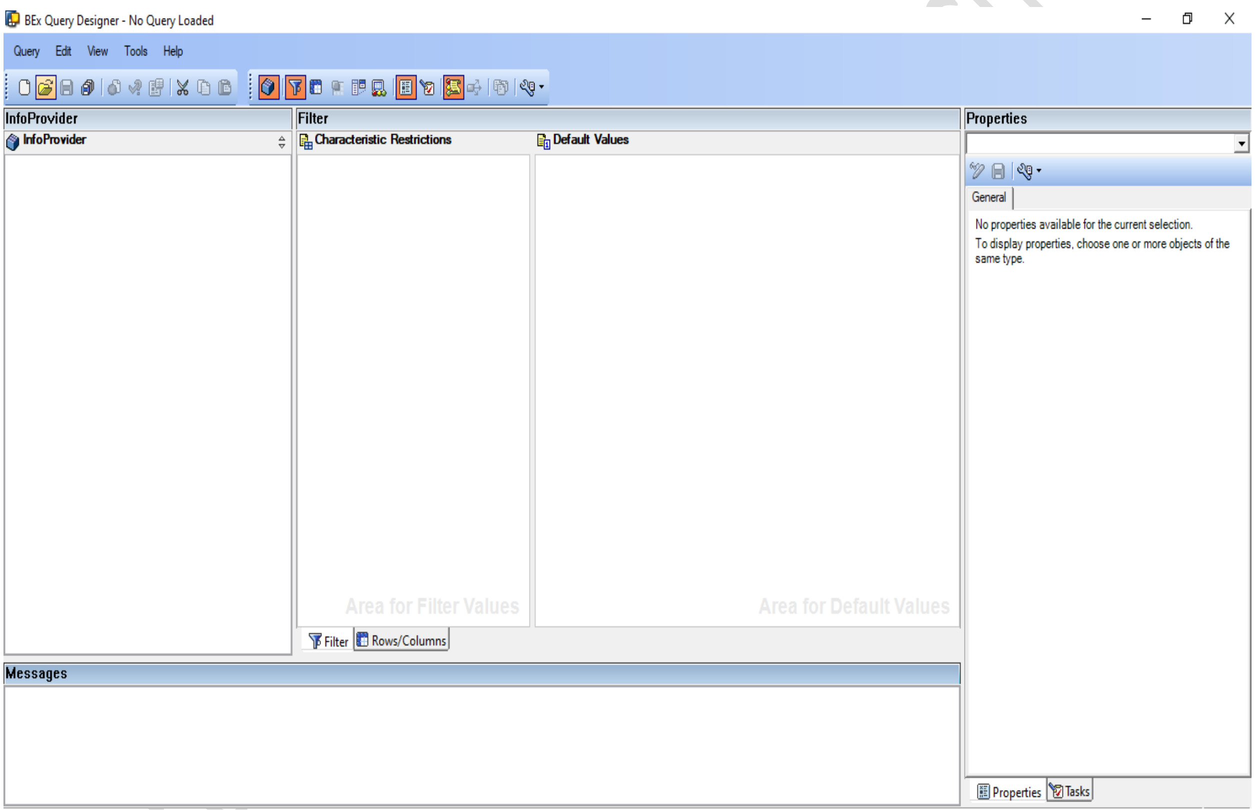
RSRT = > Start of report monitor

RRMX = > Bex Analyzer

How the query is written by using some tools like Bex(business explorer), EPM(enterprise portal management), BO(business objects)

Bex can be accessed by two components.

1) **Bex analyzer**
2) **Query designer** = > Create query/place to execute the query(RRMX)



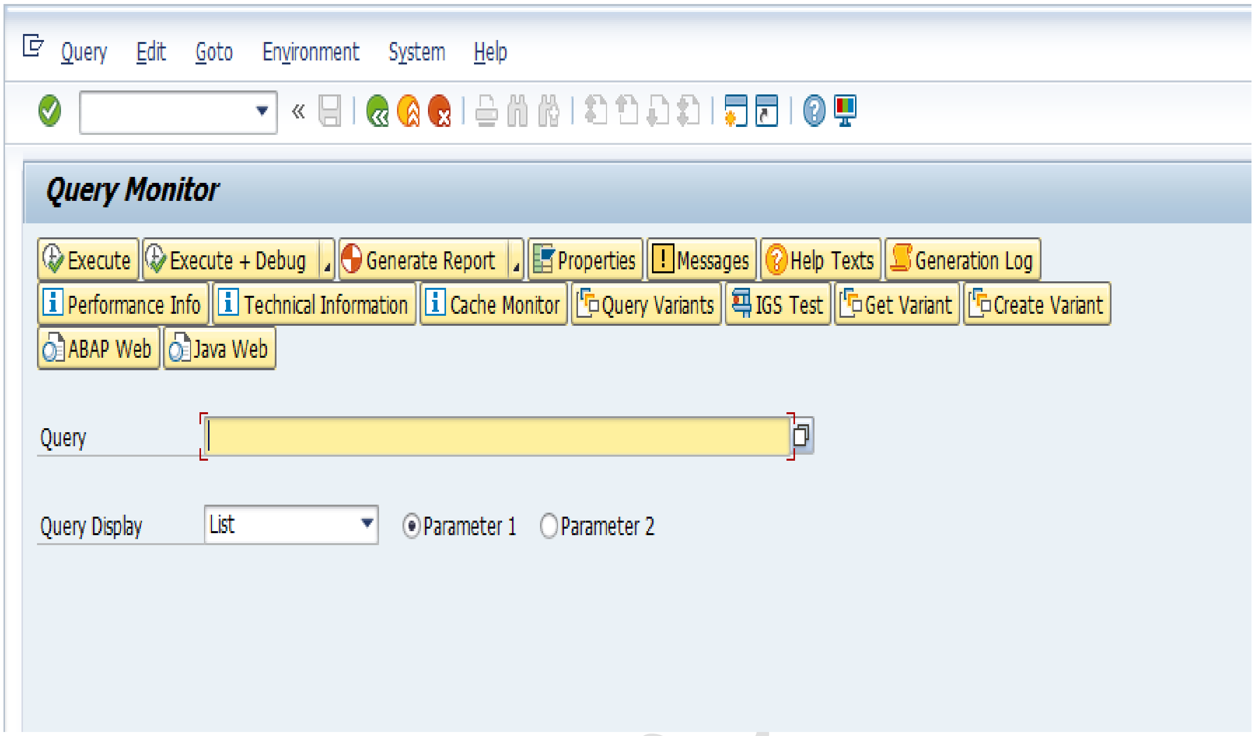
We can open the analyzer screen by using the t code called **RRMX**

RSRT for query execution.

RSRT = > query name = > search = > provide the inputs = > select data and continue = > data will display

Concentrate on technical name for ex = > ZSALES_CUBE – execute -sometime the selection screen will come enter the particular details and continue.

Below is the screenshot of **RSRT** transaction code.



How the **security** will explain here means

For ex = > **user is not able to execute the query**

- First find the **query name.**

Info area	Info cube	Queries
Benz	USA	Sales_query
		emp_data
	Germany	Sales_query
		emp_data
	India	Sales_data
		HR_data

To execute this **query(IA)** - Benz, the user need **access** to all its queries and the **containers**.

Container means, here you can see in the below picture Where IA 1 stands for info area 1. And IC 1 and 2 stands for Info cube 1 and 2 and Q1,2,3,4,5 stands for query 1,query 2,3,4,5.

IA 1			
IC 1		IC 2	
Q1			Q3
			Q4
Q2			Q5

To execute this query user, need **access** to Q1, and IC1, and IA1, to reach Q1 you need to break IC1 and IA1.

Similarly, to **execute** Q3, users need access to IC2 and IA1, and the query itself Q3.

But how the **Access controls** will be done, means by using the Authorization objects.

The Authorization Objects controlling access to queries are = >

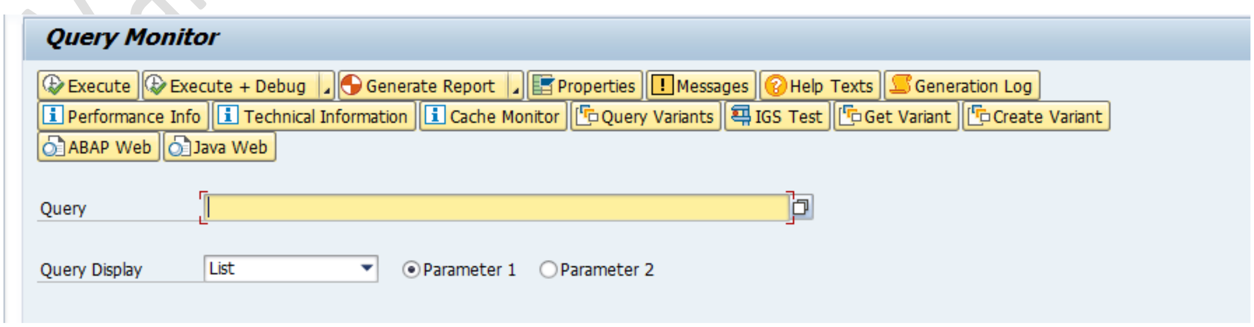
- ✓ **S_RS_COMP**
- ✓ **S_RS_COMP1**
- ✓ **S_RS_FOLD**

First find the query name, next find the info cube on which the query is written, then find the info area under which info cube is lying.

When the **query name** is known, how to find the info cube?

Query = > Sales_Z_CUBE

Go to RSRT = > **Put the Query name** = > Select the query display mode = > execute.



How do you find the **Info area** name when the **info cube** name is known?

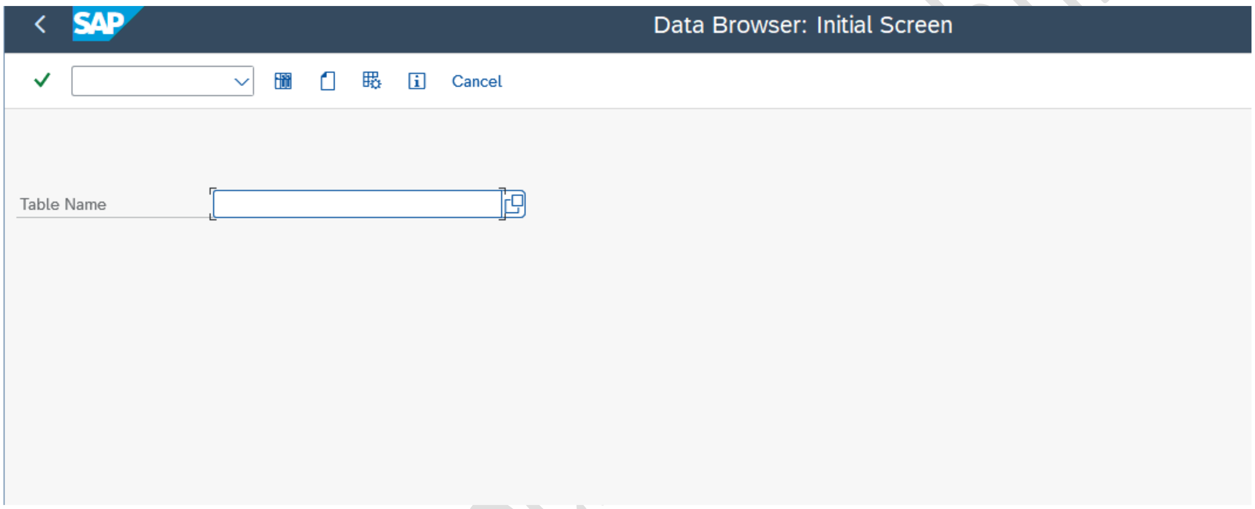
For Ex = > **Info cube** - IC_TLT

Go to **RSA1** = > Ensure Info provider is selected under **modelling** = > search = > put the info cube name and search = > find the info area name.

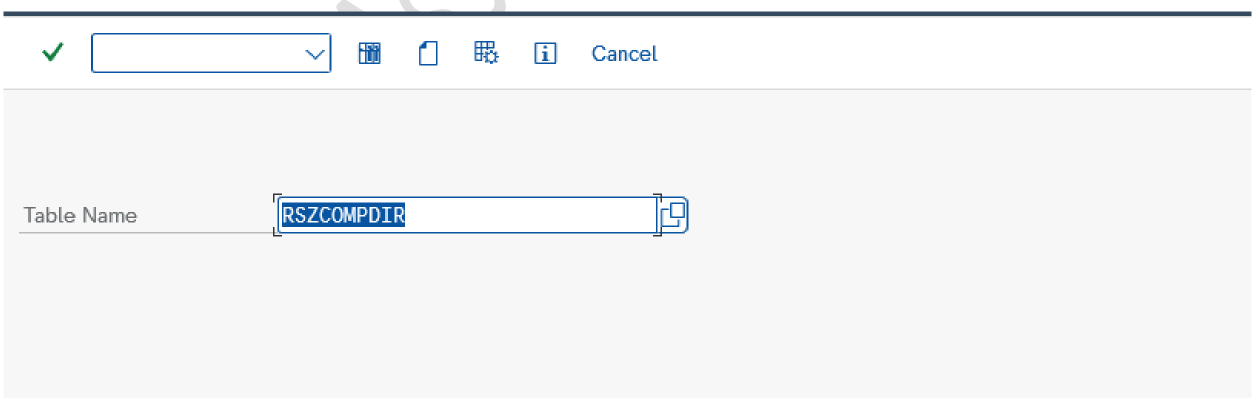
How to find the owner of a query?

Go to Table = > **RSZCOMPDIR**

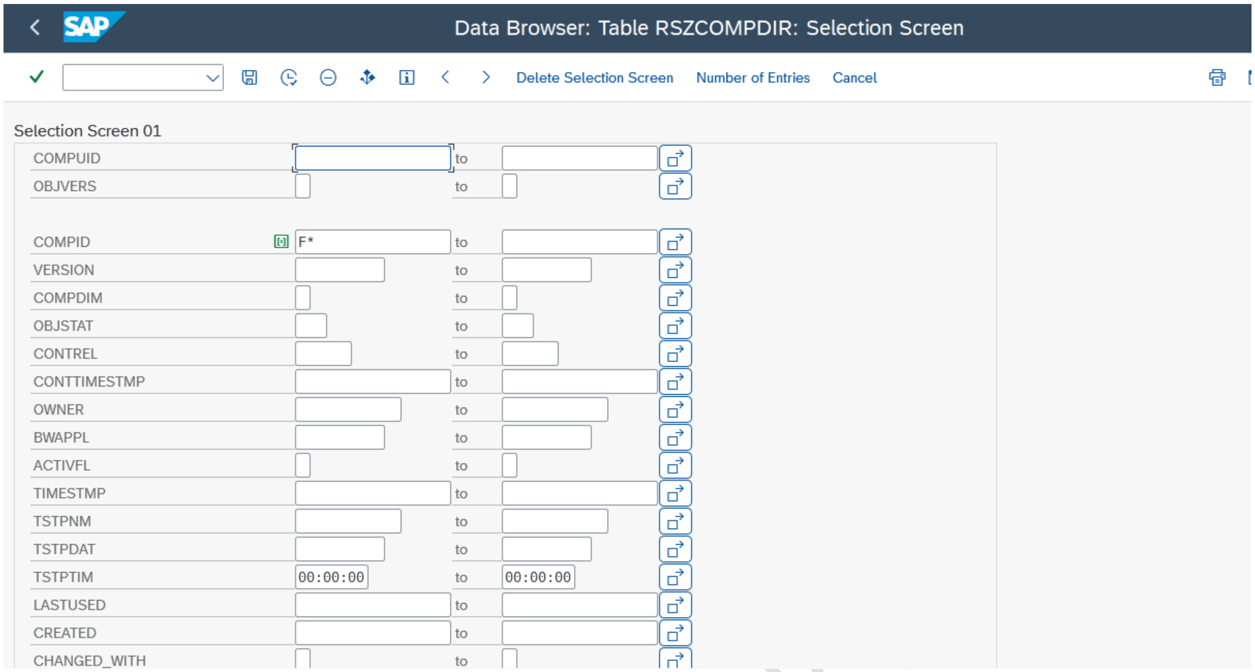
Execute = > **SE16** t code for table display



Then enter the Table name called **RSZCOMPDIR.**



Then click on enter the screen below will open then you have to give query name under **comp id and execute.**



Click on **Execute**, it will show starting with **F*** queries and **who** is the owner of that query.

COMPUID	OBJVE...	COMPID	VERSI...	COMP...	OBJST...	C...	...	OWNER	BWA...	ACTI...	TIMESTMP	TSTPNM	TSTPDAT	TSTPTIM	LASTUSED
006EIK1L...	A	FISCAL_YEAR	100		ACT		0	MONIKAH		X	20.241.224.081.809	MONIKAH	24.12.2024	08:18:09	20.250.220.104.952
006EIK1L...	M	FISCAL_YEAR	100		ACT		0	MONIKAH		X	20.241.224.081.809	MONIKAH	24.12.2024	08:18:09	20.241.224.081.809
006EIK1L...	A	FISCAL_YEAR1	100		ACT		0	MONIKAH		X	20.250.430.044.118	MONIKAH	30.04.2025	05:41:18	20.250.430.050.831
006EIK1L...	M	FISCAL_YEAR1	100		ACT		0	MONIKAH		X	20.250.430.044.118	MONIKAH	30.04.2025	05:41:18	20.250.430.044.118
006EIK1L...	A	FISCAL_YEAR2	100		ACT		0	MONIKAH		X	20.250.430.050.758	MONIKAH	30.04.2025	06:07:58	20.250.630.061.312
006EIK1L...	M	FISCAL_YEAR2	100		ACT		0	MONIKAH		X	20.250.430.050.758	MONIKAH	30.04.2025	06:07:58	20.250.430.050.758
006EIK1L...	A	FIXED_ASSET_R...	128	1	ACT		0	MONIKAH		X	20.250.623.090.458	MONIKAH	23.06.2025	10:04:58	20.250.630.061.312
006EIK1L...	M	FIXED_ASSET_R...	128	1	ACT		0	MONIKAH		X	20.250.623.090.458	MONIKAH	23.06.2025	10:04:58	20.250.623.090.458
00O2TKM...	A	FICOST_FULLYE...	11	1	ACT		0	SXMKN		X	20.150.810.151.146	SXMKN	10.08.2015	17:11:46	20.160.707.072.503
00O2TKM...	M	FICOST_FULLYE...	11	1	ACT		0	SXMKN		X	20.150.810.143.457	SXMKN	10.08.2015	16:34:57	20.150.810.143.457
00O2TKM...	A	FICOST_QTR	11	1	ACT		0	SXMKN		X	20.150.810.151.150	SXMKN	10.08.2015	17:11:50	20.150.821.145.600
00O2TKM...	M	FICOST_QTR	11	1	ACT		0	SXMKN		X	20.150.810.145.031	SXMKN	10.08.2015	16:50:31	20.150.810.145.031
00O2TKM...	A	FICOST_PERQTR...	11	1	ACT		0	SXMKN		X	20.150.730.150.628	SXMKN	30.07.2015	17:06:28	20.150.821.132.558
00O2TKM...	M	FICOST_PERQTR...	11	1	ACT		0	SXMKN		X	20.150.123.152.013	SXMKN	23.01.2015	16:20:13	20.150.123.153.338
00O2TKM...	A	FICOST_3MNTH_...	11	1	ACT		0	SXMKN		X	20.150.807.155.118	SXMKN	07.08.2015	17:51:18	20.150.821.132.558
00O2TKM...	M	FICOST_3MNTH_...	11	1	ACT		0	SXMKN		X	20.141.024.090.547	SXMKN	24.10.2014	11:05:47	20.141.027.122.855
00O2TKM...	A	FICOST_YTD_SUM	11	1	ACT		0	SXMKN		X	20.150.730.150.629	SXMKN	30.07.2015	17:06:29	20.150.821.132.557
00O2TKM...	M	FICOST_YTD_SUM	11	1	ACT		0	SXMKN		X	20.141.024.092.742	SXMKN	24.10.2014	11:27:42	20.141.027.122.454

Most important authorization objects and some more information in PART 2

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